conversion mach d0.8 — values and first derivatives

prescribed reference
current implementation

Y
Outlet row-frame Mach number

d(Y) / dx
near constant: expanded display span
Outlet row-frame Mach number

Y flag false
Outlet row-frame Mach number

d(Y flag false) / dx
Outlet row-frame Mach number

Y formula unrestricted
Outlet row-frame Mach number

d(Y formula unrestricted) / dx
Outlet row-frame Mach number

raw roundtrip
roundtrip display span expanded; max |value - input| = 2.2e-16
Outlet row-frame Mach number

d(raw roundtrip) / dx
roundtrip display span expanded; max |d/dx| = 4.5e-11
Outlet row-frame Mach number

roundtrip delta phi2
Outlet row-frame Mach number

d(roundtrip delta phi2) / dx
Outlet row-frame Mach number

input delta phi2
Outlet row-frame Mach number

d(input delta phi2) / dx
near constant: expanded display span
Outlet row-frame Mach number